Forecasting task

$\mathbf{x} = (x_{t-T+1}, \ldots, x_t) \in \mathbb{R}^T$

$\mathbf{y} = (x_{t+1}, \ldots, x_{t+H}) \in \mathbb{R}^H$

T: context length; H: forecast horizon.

$$\mathbf{x} \longmapsto p_\theta(\mathbf{y} \mid \mathbf{x})$$

Quantiles and intervals describe forecast uncertainty.

Example: 96 past hours, a 24-hour forecast.

How DecoVAE-FM represents uncertainty

Trend and seasonal outputs form the mean. Residual features also inform the variance.

$$\mu = \tau + s, \qquad \log \sigma^2 = g_\theta(\mathbf{z})$$

τ, s: trend / seasonal forecasts.

$\mathbf{z}$ collects learned feature vectors of the trend, seasonal and residual components, obtained through their encoders.

Mean and variance are in normalized model space.